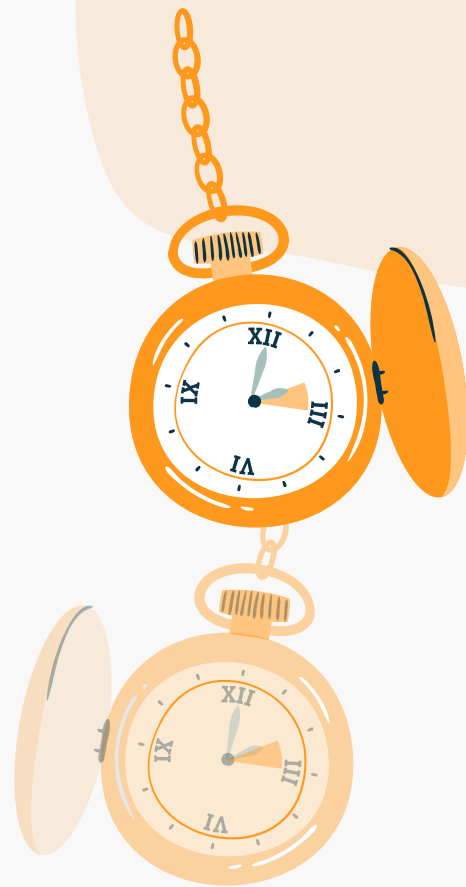


Parallelizing Image Color Quantization:

Accelerating K-Means using
OpenMP, MPI, and CUDA

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Motivation & Background

Motivation: The Cost of Color Reduction

- **Goal:** Reduce the number of distinct colors in an image (Color Quantization).
- **Algorithm: K-Means Clustering**
 - Iteratively finds K dominant centroids.
 - Reassigns every pixel to the nearest centroid.
- **The Problem:** High Computational Complexity
 - Complexity $\approx O(N_{pixels} \times K_{clusters} \times Iterations)$
 - **Bottleneck:** Extremely slow for high-resolution images on serial processors.



Project Goal

Project Goal: Achieving Parallel Speedup

Objective: Implement and benchmark **three distinct parallel architectures** against a Serial Baseline to evaluate speedup.

Multi-core CPU	Distributed Cluster	GPU
		



Acceleration Techniques

OpenMP

- Loop Parallelism: Uses **collapse(2)** to flatten nested loops and improve core utilization.
- Lock-Free Reduction: Uses **reduction(+: ...)** for lock-free summation.

MPI

- Data Decomposition: Uses **MPI_Scatterv** to distribute image data across nodes.
- Distributed Reduction: Uses **MPI_Allreduce** to synchronize local sums into global centroids.
- Broadcast: Uses **MPI_Bcast** to share updated centroid coordinates across all ranks.

Acceleration Techniques

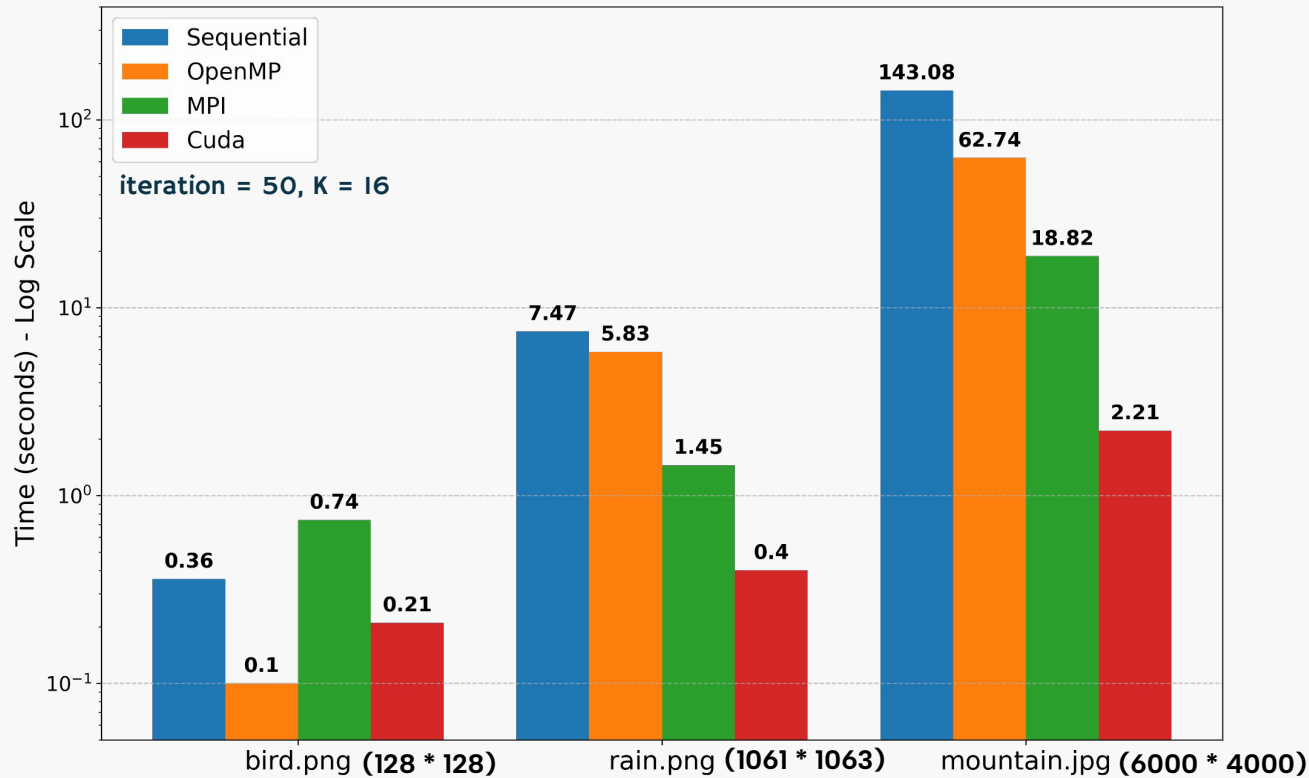
CUDA

- **Constant Memory:** Caches k centroids in `__constant__ memory` for fast broadcast access.
- **Shared Memory:** Uses `__shared__ memory` for block-level reduction to reduce global memory latency.
- **Full GPU Execution:** Keeps the **entire K-Means iterative training loop** (both the pixel assignment phase and the centroid update phase) **on the GPU** to eliminate slow CPU-GPU (PCIe) transfers.



Evaluation Results

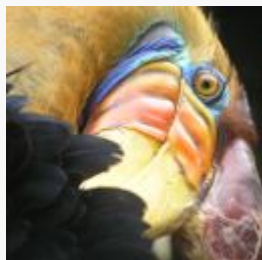
Performance of Different Methods



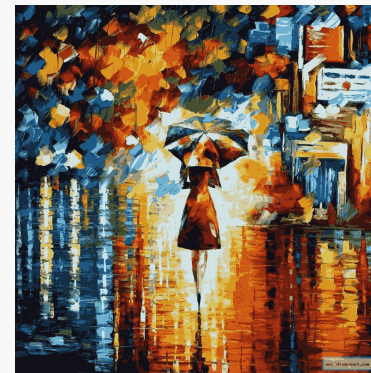
Visual Demonstration

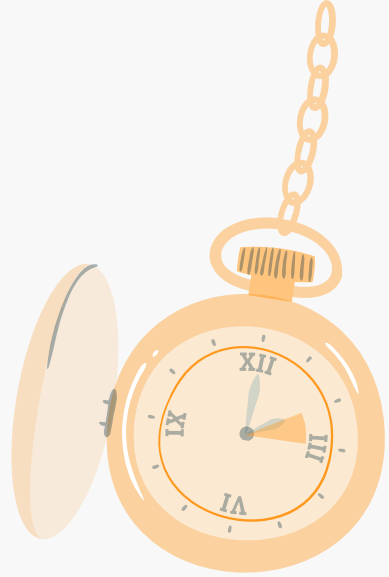
Rain.png (1061 * 1063)

Bird.png (128 * 128)



Mountain.jpg (6000 * 4000)





Thanks!



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